

AIMS Lab Research Engineer Selection Assignment

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Task 1: Scientific review

Article. Petkeviciene J, Klumbiene J, Kriaucioniene V, Raskiliene A, Sakyte E, and Ceponiene I. “Anthropometric measurements in childhood and prediction of cardiovascular risk factors in adulthood: Kaunas cardiovascular risk cohort study.” *BMC Public Health*. 2015;15:218. doi:10.1186/s12889-015-1528-5.

Purpose and design. Childhood obesity is a growing public-health problem, but it is uncertain whether early adiposity has an independent effect or primarily reflects the tracking of excess weight into adulthood. The study therefore asked whether adiposity at ages 12–13 predicts cardiovascular risk at ages 48–49 independently of weight gained in the intervening 35 years. In 1977, 1,082 randomly sampled sixth-grade pupils from 15 schools in Kaunas, Lithuania, underwent standardized measurement of height, weight, blood pressure, and triceps and subscapular skinfolds. In 2012, 507 of 794 eligible survivors participated (63.9%); 506 had adequate measurements for analysis. Adult assessment included body mass index (BMI), waist circumference, blood pressure, glucose, lipids, and high-sensitivity C-reactive protein (CRP), with questionnaire data on smoking, alcohol, physical activity, and family history of obesity.

Methods. Spearman correlations quantified tracking of childhood and adult anthropometry. Separate logistic regressions related childhood BMI or the sum of skinfolds to adult obesity, central obesity, metabolic syndrome, hyperglycaemia/diabetes, hypertension, dyslipidaemia, and elevated CRP. Models adjusted for sex, adult lifestyle variables, family history, and, where stated, BMI gain. This approach is suitable for binary outcomes, although testing many outcomes without a reported multiplicity correction increases the chance of false-positive findings.

Baseline anthropometry (1977, ages 12–13) → adult risk factors (2012, ages 48–49) → adjusted logistic models

Principal findings. Weight and BMI tracked moderately from childhood to adulthood. Weight correlations were $r = 0.45$ in men and $r = 0.56$ in women; BMI correlations were $r = 0.41$ and $r = 0.53$, respectively (all $p < 0.001$). Adult obesity rose sharply across childhood-BMI quintiles: in the highest quintile, 50.0% of men and 56.4% of women were obese at follow-up. After adjustment for BMI gain, each 1-kg/m² higher childhood BMI was associated with metabolic syndrome (odds ratio [OR] 1.21, 95% CI 1.10–1.32), hyperglycaemia/diabetes (OR 1.15, 1.06–1.24), and elevated CRP (OR 1.19, 1.09–1.31). Childhood BMI was not associated with adult hypertension, raised triglycerides, or low HDL cholesterol; BMI gain was associated with all three (ORs 1.26, 1.22, and 1.20 per kg/m², respectively). Childhood skinfolds showed a broadly similar pattern.

Critical appraisal. Major strengths are the unusually long follow-up, random school sampling, direct standardized measurements at both examinations, and consideration of several behavioral confounders. Baseline anthropometry did not differ detectably between participants and nonparticipants. Important limitations remain: only 46.8% of the original cohort entered the final analysis, so selective survival, migration, and nonresponse may bias results; the single-city, single-birth-year sample limits generalizability; lifestyle was not measured comparably across the life course; and residual socioeconomic, dietary, and developmental confounding is plausible. Childhood BMI and BMI gain are mathematically coupled to adult BMI, producing collinearity and, for adult obesity, complete separation. The observational design supports prediction and association, not causation.

Conclusion. The evidence supports two complementary prevention messages: higher childhood adiposity identifies people at elevated later metabolic risk, while excessive BMI gain across life is particularly relevant to hypertension and dyslipidaemia. Maintaining healthy weight should therefore begin in childhood and continue through adulthood. Replication in larger, multi-site cohorts with repeated lifestyle measures and prespecified outcome correction would strengthen causal and population-level interpretation.

Task 2: Statistical analysis of the supplied dataset

Question, data preparation, and algorithm

The analysis asks: *which available non-identifying features are associated with the dataset's recorded hypertension flag?* The file contains 29,999 individuals from 21,449 households; 1,575 (5.25%) are marked hypertensive. Names, dates, and household/user identifiers were excluded. Clinically interpretable categories were used: age (0–17, 18–39, 40–59, ≥ 60 years), the supplied income and status categories, and collapsed high-severity levels for blood pressure, glucose, and oxygen saturation. BMI was dichotomized as overweight/obese versus lower categories to satisfy expected-count requirements. Each test used complete cases for that feature.

For a feature with r levels and a binary outcome, the Pearson chi-square statistic is

$$E_{ij} = \frac{R_i C_j}{N}, \quad \chi^2 = \sum_{i=1}^r \sum_{j=1}^2 \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \quad df = (r - 1)(2 - 1).$$

Here O_{ij} is the observed cell count and E_{ij} is the count expected under independence. A small p -value rejects independence but does not show causality. Effect size was reported as Cramer's $V = \sqrt{\chi^2 / [N \min(r - 1, 1)]}$; roughly, 0.10, 0.30, and 0.50 denote small, medium, and large associations. Benjamini–Hochberg (BH) adjustment controlled the false-discovery rate across the ten Pearson tests. The prespecified significance rule was $p_{\text{BH}} < 0.05$. Expected-count adequacy was checked; sparse binary tables were assessed separately with Fisher's exact test.

Results

Feature	Complete N	$\chi^2(df)$	p_{BH}	Cramer's V	Interpretation
Recorded diabetes	29,999	3439.6 (1)	< .001	.339	58.1% vs 4.2% hypertensive
Union/location	29,999	1611.1 (15)	< .001	.232	prevalence range 0–24.3%
Age group	29,999	1242.9 (3)	< .001	.204	13.6% at ≥ 60 ; 1.7% at 18–39
Current BP category	27,600	1065.3 (3)	< .001	.196	12.1% in combined high group
Glucose-status category	1,583	238.9 (3)	< .001	.388	strong, but coding depends on diabetes
Pulse category	27,455	29.35 (2)	< .001	.033	statistically significant; negligible size
Income class	29,999	9.96 (3)	.027	.018	statistically significant; negligible size
Overweight/obese BMI	1,128	4.57 (1)	.041	.064	small association; 96.2% missing
Oxygen-saturation category	4,345	4.05 (1)	.049	.031	marginal significance; negligible size
Sex	29,999	2.22 (1)	.136	.009	not significant

The strongest usable associations were diabetes, location, age, and current BP category. Diabetes had both a very small adjusted p -value and a meaningful effect size: hypertension prevalence was 58.1% among 601 people with recorded diabetes versus 4.2% among those without it. Hypertension also increased substantially with age. Geographic variation was large, but it may reflect different enrollment, ascertainment, or coding practices across unions rather than a biological location effect. The current-BP association is expected and is not an independent risk factor because blood pressure is part of the hypertension construct.

Pulse, income, BMI, and oxygen saturation passed the significance threshold but had $V < 0.07$; their practical associations are small or negligible. The BMI result is especially uncertain because only 3.8% had measurements. The oxygen result was counterintuitive (7.3% hypertension with normal saturation versus 3.8% with low saturation), suggesting selection or measurement bias. Sex was not significant. Glucose status appears strong, but the labels are conditionally coded: categories such as “High” occur primarily among people already marked diabetic. It should not be treated as an independent predictor without recoding raw glucose by fasting/random condition.

Stroke history (23 cases), CVD history (34 cases), and disability (69 cases) failed Pearson expected-count criteria. Exploratory Fisher tests nevertheless found higher odds of recorded hypertension (OR 65.7, 33.5, and 4.63, respectively; all $p < .001$). These estimates are unstable, potentially co-recorded diagnoses, and should not be used for prediction without more cases.

Secondary outcome check. Diabetes had enough positive records ($n = 601$) for a separate screen. Excluding the diagnosis-dependent glucose label, its largest valid associations were union ($\chi^2 = 652.38$, $V = .147$), age group ($\chi^2 = 366.96$, $V = .111$), and current BP category ($\chi^2 = 122.07$, $V = .067$; all $p < .001$). Income was also significant but small ($\chi^2 = 38.43$, $V = .036$), whereas sex was not ($p = .257$). Stroke and CVD have only 23 and 34 positive records, respectively, so broad multi-category Pearson screening is unreliable; exact or penalized regression methods are preferable.

Data limitations and recommended analytic outcomes

Missingness is highly uneven: blood pressure and pulse are about 92% observed, but height/weight/BMI only 3.8%, glucose 5.3%, oxygen saturation 14.5%, and MUAC 0.2%. The poverty flag is constant, six people are freedom fighters, 140 ages exceed 100 years (134 equal 120), and 14,973 records share a household with another participant. Consequently, complete-case comparisons may be selected, implausible ages require verification, and ordinary tests understate uncertainty from household clustering. The dataset is cross-sectional and does not contain the paired childhood/adult measurements, skinfolds, lipids, CRP, or BMI gain needed to reproduce the Kaunas longitudinal analysis.

The most defensible next outcome is an explanatory multilevel logistic model for recorded hypertension using verified age, sex, diabetes, income, and union, with a household random intercept or cluster-robust standard errors. BP category and diagnosis-derived glucose labels should be omitted from a forward-looking model to prevent target leakage. A second analysis could model elevated measured BP among adults using age, sex, diabetes, income, and location. Report adjusted ORs with 95% confidence intervals, calibration, and sensitivity analyses for missing data and alternative age cleaning. Prospective cardiovascular-risk prediction or causal claims are not supported until longitudinal outcomes and consistent clinical measurements are collected.

Operationally, field teams should standardize collection of height, weight, glucose condition, oxygen saturation, and MUAC; audit missingness by union; and confirm that apparent geographic hotspots persist after age adjustment and measurement-quality checks before reallocating screening resources.